

PRAGMATISM

Build Things People Use

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PLAN ₿ SUMMER SCHOOL · LUGANO

CLOSING KEYNOTE · 2026

For Both Tracks · Business & Developer

One talk. **Two** tracks.

BUSINESS TRACK

Identify the market.

Stablecoins, the regulatory turn, how merchants actually adopt, the opportunity you're about to go work in.

DEVELOPER TRACK

Build the tech.

HTLCs, atomic routing, gas abstraction, what shipping real, non-custodial infrastructure actually looks like.

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PART ONE

**The soundest money, yet
barely used.**

//

**Make something
people want.**

The only mandate that matters.

The Win · 17 Years Live

Bitcoin is the
cypherpunk idea
that **actually shipped.**

Seventeen years. A new block every ten minutes. No trusted third party. We won...?

Code nobody uses
is just code.

The same way a Bitcoin node you're running but never connect to your wallet
is just... a curiosity.

Money nobody uses
for daily spending
is... not money.

We built the soundest money in history, but it fails to change ordinary people's lives if one can't spend it.

We won the protocol. We did not win **daily life**.

LIGHTNING NETWORK · NOV 2025

\$1.17_B/mo

Best month on record. **5.2M payments**, avg \$223, mostly exchange flow, not coffee.

River · Sam Wouters, Feb 2026

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vs.

VISA + MASTERCARD · MONTHLY

\$2.28T/mo

Combined settlement, **every single month** · 7–9% YoY.

Visa & Mastercard FY2025 reports

02

PART TWO

**Meanwhile, the dollar got an
"API" via Stablecoins**

Calling It What It Is

Stablecoins are **Fiat**. Full stop.

A wrapped dollar IOU. Private issuer. Compliant (Read: "freezable").

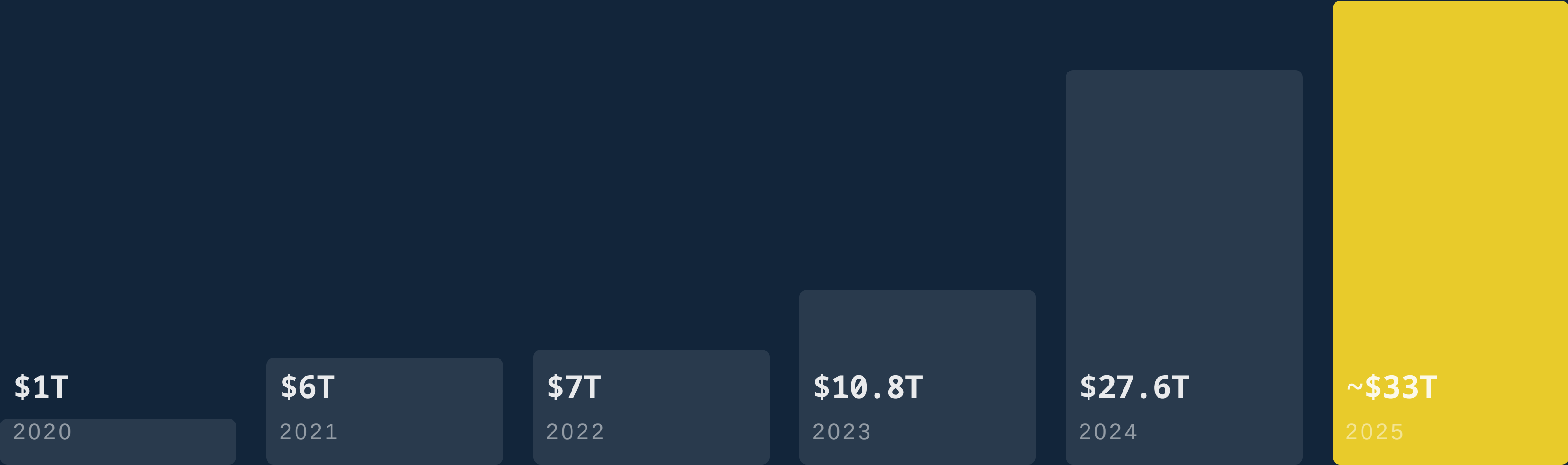
Stablecoin Supply · Jan 1 · 2020 → 2026

From \$4B to over \$300B, while we weren't paying attention.

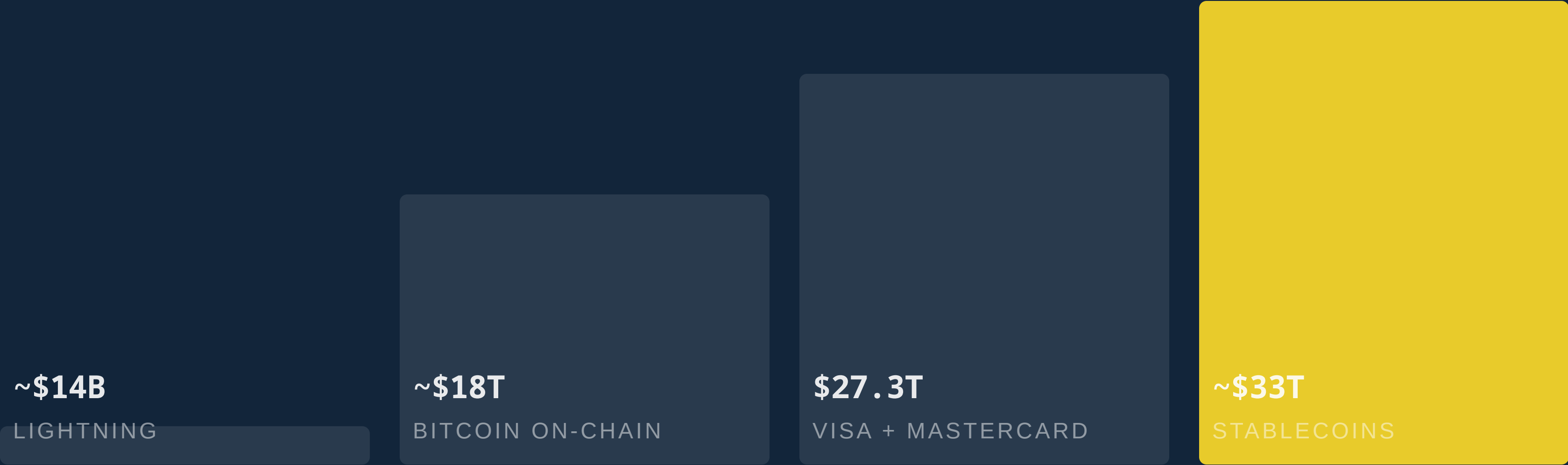


In 2026 USDT & USDC accounted for ≈ 80% of supply

From \$1T to \$33T while we argued Covenant OP Codes.



Who paid attention?



Washington just gave dollar tokens a **home**.

2025 · ENACTED

GENIUS Act

Federal framework for **payment stablecoins**.

Issuer licensing. 1:1 reserves in cash & T-bills. Disclosure rules. Stablecoins are now first-class US payment instruments.

2026 · PENDING

CLARITY Act

SEC vs. CFTC, the line being **drawn**.

Would split securities from digital commodities, so issuers, exchanges and DeFi know which agency owns which token. Passed the House in 2025; awaiting the Senate.

TETHER'S RESPONSE

USA ₮

A US-domiciled stablecoin **built for GENIUS**.

Launched Jan 2026. Issued by Anchorage Digital Bank, reserves at Cantor Fitzgerald. Tether's move to onshore its US business, and take Circle on home turf.

One law on the books, one on the way: the US is finally writing crypto's rulebook.

Stablecoins are going **local**.

EXHIBIT A · 25 MAY 2026



GEL ₾



Georgian Lari stablecoin

Issued by Tether, jointly developed with the Government of Georgia.

National-currency stablecoins now have government partners.

\$0 → \$400M /MO

BRLA (Brazilian-Real stablecoin) monthly transfer volume from early 2023 to early 2026.

50% → 75% DOMESTIC

Share of stablecoin payments that stay inside one country from 2024 to 2026. Less cross-border remittance, more everyday local settlement.

The agentic web is starting to emerge. And it pays for itself. **Just not in sats.**

L402 · LIGHTNING · 2020

690 endpoints

30 providers. Lightning leaves no chain footprint. The integration count after 6 years is the signal.

402index.io · 27 May 2026

VS.

X402 · COINBASE · MAY 2025

44.6k endpoints

604 providers. 65× L402 in 12 months.

402index.io · 27 May 2026

Same HTTP 402 spec. Lightning is technically the best at micropayments but the agents went elsewhere.

The Honest Disclosure

Regulated & Freezable. By design.

The Superpower

Open to anyone. By design.

No bank approval. No paperwork. No permission. A dollar that anyone on Earth can hold, receive and send, the same way anyone can run a Bitcoin node. That openness is the whole point.

THIS IS THE PART NOBODY TALKS ABOUT

...they settle on rails we can natively interact with.



USDT

Tether · global

~\$185B supply



USDC

Circle · US

~\$76B supply



USAT 

Tether · US-domiciled

Launched Jan 2026 ·
GENIUS-compliant



EURC

Circle · Euro

MiCA-compliant · EU

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PART THREE

How do we make stablecoins work **for** Bitcoin?

The Merchant Side

Merchants don't want Bitcoin. They want **dollars**.

THE MERCHANT ASK

**Predictable accounting.
Instant settlement. No FX
risk.**

Bitcoin volatility kills the argument. Holding Bitcoin means headache and perceived "risk". Asking a merchant to take Bitcoin loses the sale.

THE PRAGMATIC MOVE

**Give them the dollar. Settle
Bitcoin behind the scenes.**

The merchant never sees or touches Bitcoin. The customer never holds stablecoins. Everyone gets what they came for.

You don't have to imagine it. It's the **street outside.**

360+

Lugano merchants accept Bitcoin, USD₣ & LVGA, cafés, boutiques, even McDonald's.

5_{sec}

Lightning settlement at the counter. The merchant keeps Swiss francs if they want.

LVGA = CHF

The city itself takes BTC & USDT for every tax bill, no upper limit, from childcare to naturalisation fees.

The customer scans sats. The merchant keeps francs. The highest-density Bitcoin-and-stablecoin payment city on Earth, and you've been standing in it for two weeks.

THE USER SIDE

Bitcoiners don't want dollars. They want to **spend** their Bitcoin.

USE CASES · LIVE TODAY

Top up a crypto debit card.
Deposit into your neobank.
Pay a merchant.

Without leaving your Bitcoin wallet. Without giving up custody. Without an account. Sats ↔ stable, on demand.

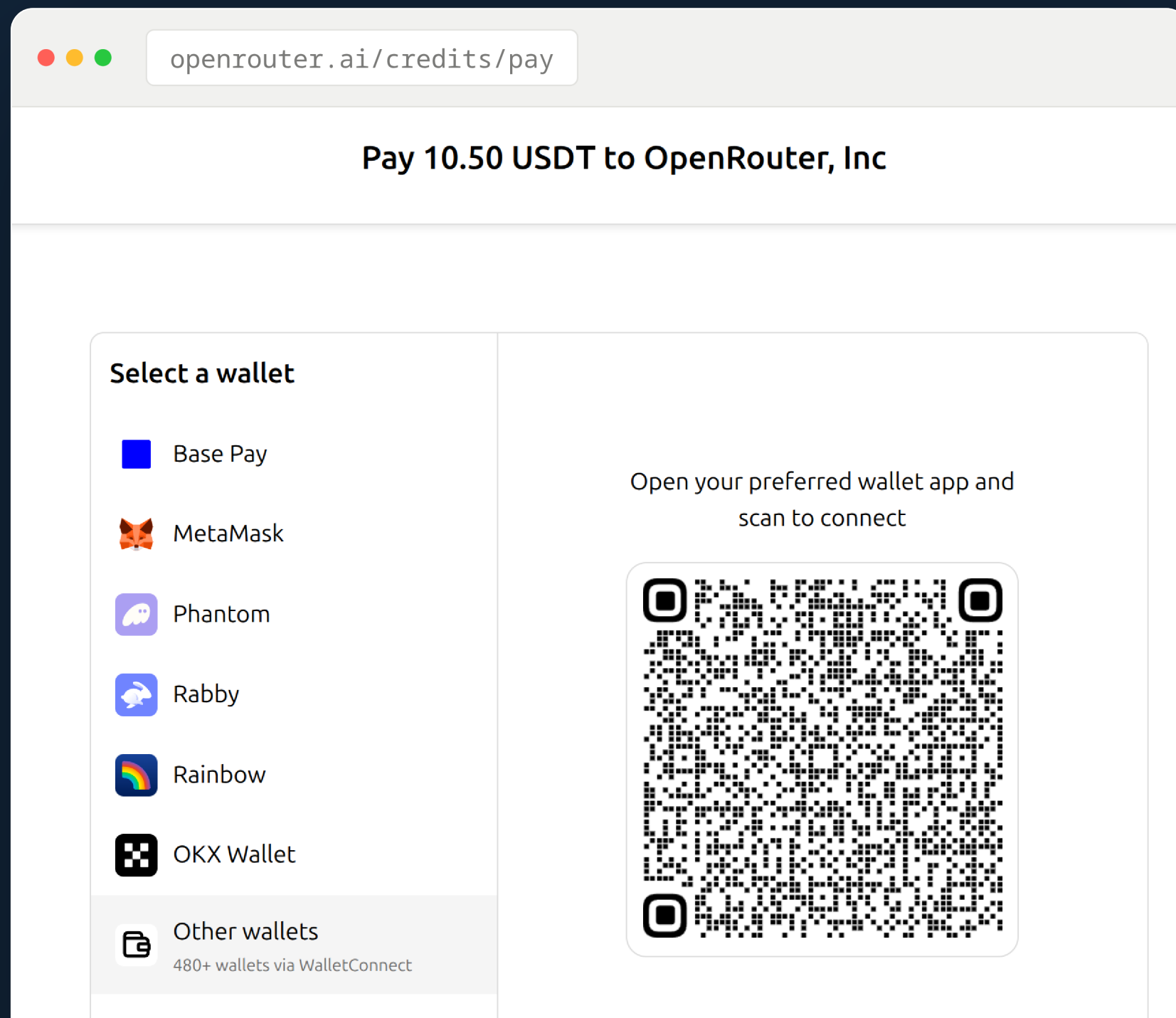
THE HUGHES TEST

Selectively reveal yourself to the Fiat system, only when you have to.

Swap to stablecoins, the new fiat, in the moment when you need to. Keep your privacy. The 1993 manifesto approves.

A REAL USDT CHECKOUT · TODAY

A typical merchant checkout. It only takes **USDT**.



WHAT THIS CHECKOUT ASKS FOR

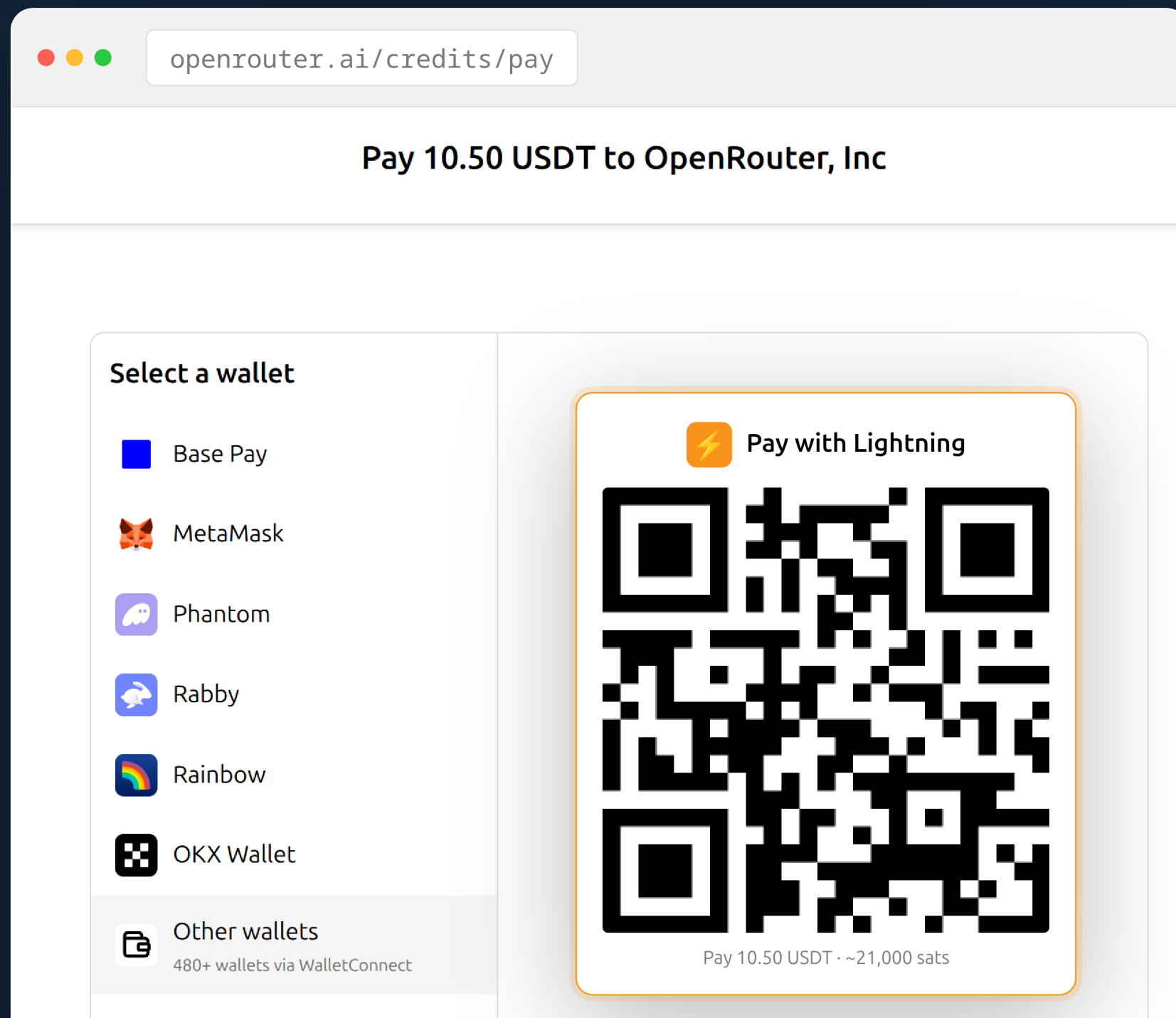
A **USDT** balance.
On the right chain.
In a wallet they recognize.

MetaMask, Phantom, Rabby, OKX, WalletConnect, all things many Bitcoiners are not very familiar with.

No Lightning. No Bitcoin. No sats.

The Move · Make Bitcoin Compatible With Any USDT Checkout

Slap Lightning on top. Same merchant. Same **USDT**.



SAME CHECKOUT. LIGHTNING ADDED.

You paid **sats**.
Merchant settles in USDT.

Drop Lightning into any USDT or USDC checkout. Without the merchant touching Bitcoin, and without Bitcoiners touching USDT.
And most importantly: Without adding trust assumptions.

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PART FOUR

In **cryptography** we trust. No added trust assumptions.

The Principle · Atomic Settlement

One signature. One transaction. **Atomic.**

Either everything settles, or nothing happened. You never give up control over your funds and there is no room for a middleman to rug.



The Hard-Won Lesson

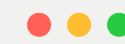
We tried the obvious way. It failed miserably.

A naïve Lightning ↔ USDT HTLC hands one side a **free option**: Bitcoin moves against the dollar while the swap is pending, and whoever holds the preimage just walks away if the rate turned against them. Not theoretical, a known, real attack whenever a single HTLC spans two different assets.

THE LESSON

Failing in public is the **method**.

In 2019 the free options problem drained funds from Boltz. We got rekt. So we wrote down exactly how we failed, and fixed it. The swap engine you just saw exists **because** we failed first. Trial and error isn't a flaw in a startup, it's how you ship.



blog.boltz.exchange/p/the-problem-with-free-options

BOLTZ'S BLOG

The Problem with Free Options

TL;DR: there might not be any such thing as a free lunch, but in cryptoverse there's something called the *free options problem* that pushed us into a real quandary...



Boltz · Jul 26, 2019

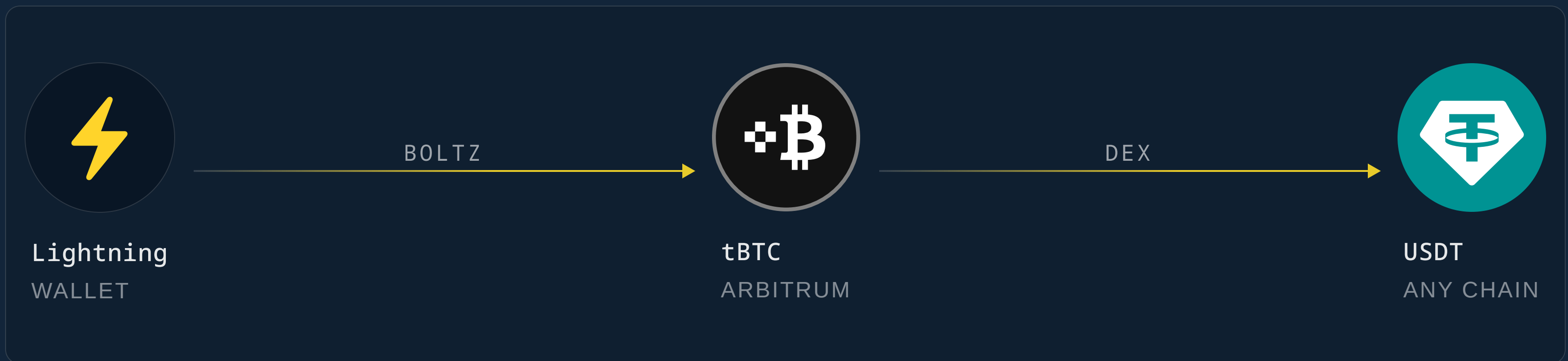
Simply put...

WE GOT REKT

THE ROUTED SWAP · HOW THE LIGHTNING SETTLES TO USDT

Lightning → tBTC → USDT. In **one** transaction.

AND ALL OF THIS RUNS CLIENT-SIDE, E.G. IN YOUR BROWSER.



claim

Boltz releases tBTC against your Lightning preimage.

execute

Router swaps tBTC → USDT on a DEX. Slippage-protected.

sweep

USDT lands in your wallet, Ethereum, Arbitrum, Optimism, Polygon, Rootstock, any USDT0 chain.

All three steps run in one transaction. If any leg fails, the whole thing reverts.

Two Building Blocks

Two pieces make the route **possible**.



Bitcoin as an **open** ERC-20. Like WBTC, but anyone can mint, unmint and use it. No gatekeeper. That's how Lightning reaches the Ethereum world without trusting a custodian.



Omnichain Tether on LayerZero's OFT standard. **One** liquidity pool on Arbitrum serves Ethereum, Optimism, Polygon, Solana, Tron. USDC reaches every chain the same way via Circle's CCTP.

USDT0 · LayerZero OFT · One Pool, Every Chain

Concentrate liquidity once. Serve **every** chain.



USDT0

LIQUIDITY HUB · ARBITRUM

LAYERZERO

Ethereum

Optimism

Polygon

Rootstock

Solana

Tron

USDC reaches the same chains via Circle's CCTP

One pool on Arbitrum. Every connected chain served. No liquidity fragmentation.

Claim. Execute. Sweep. One transaction.

```
// Router.sol, simplified
function claimERC20Execute(...) external {
    // 1 · claim tBTC with your Lightning preimage
    claimERC20Swap(claim);

    // 2 · swap tBTC → stablecoin on a DEX
    executeCalls(calls);

    // 3 · send it out, revert if below minAmountOut
    sweep(destination, token, minAmountOut);
}
```

One EIP-712 signature authorises the whole claim-and-execute. If any line reverts, the entire transaction reverts, there is no half-finished swap, ever. That atomicity is what removes the middleman.

You see the real rate **before** you sign.

real-time

A live quote API prices the exact tBTC ↔ stablecoin pair on Arbitrum for your exact amount.

minAmountOut

An on-chain floor. If the DEX returns less than this, the whole transaction reverts. No surprise haircuts.

re-approve

If the rate drifts past tolerance before you claim, the swap pauses and asks you to approve the new price.

The Execution Layer + The Proof

Why **Arbitrum**.

BEST TOOL FOR THE JOB

Low gas. Cheap enough to run multi-leg swaps profitably.

Fast OFT exit. USDT0 bridges out quickly to any chain.

Deep liquidity. Real markets for both tBTC and stablecoins.

OPEN SOURCE FTW

The best validation of a sound design is others building on it. Our Router contract and the tBTC-on-Arbitrum pattern we pioneered are **already adopted** by other teams. It's all open, read it, fork it, ship your own.

What Is **Not** Required

It should feel like Bitcoin. So it does.

// NO

ETH.

Gas abstraction handles gas on your behalf.
You never hold an ETH or other gas balance.

// NO

KYC.

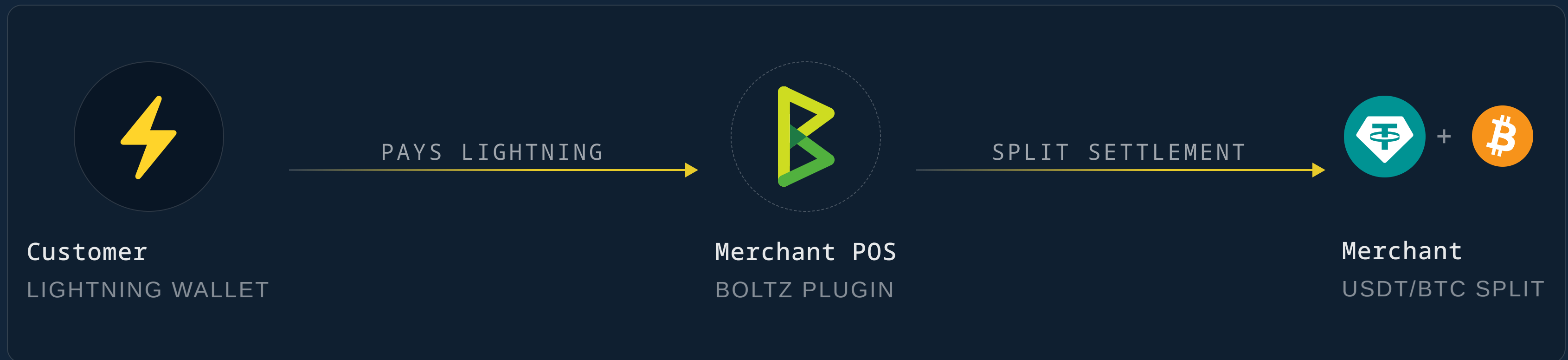
No documents. No selfie. Nobody needs to
know your name.

// NO

Accounts.

No email. No password. No Sign-Up.

Accept Lightning. Settle in **dollars**. Or split BTC and USDT.



05

PART FIVE

**Be pragmatic. Build for the
real world.**

Adoption is the war.

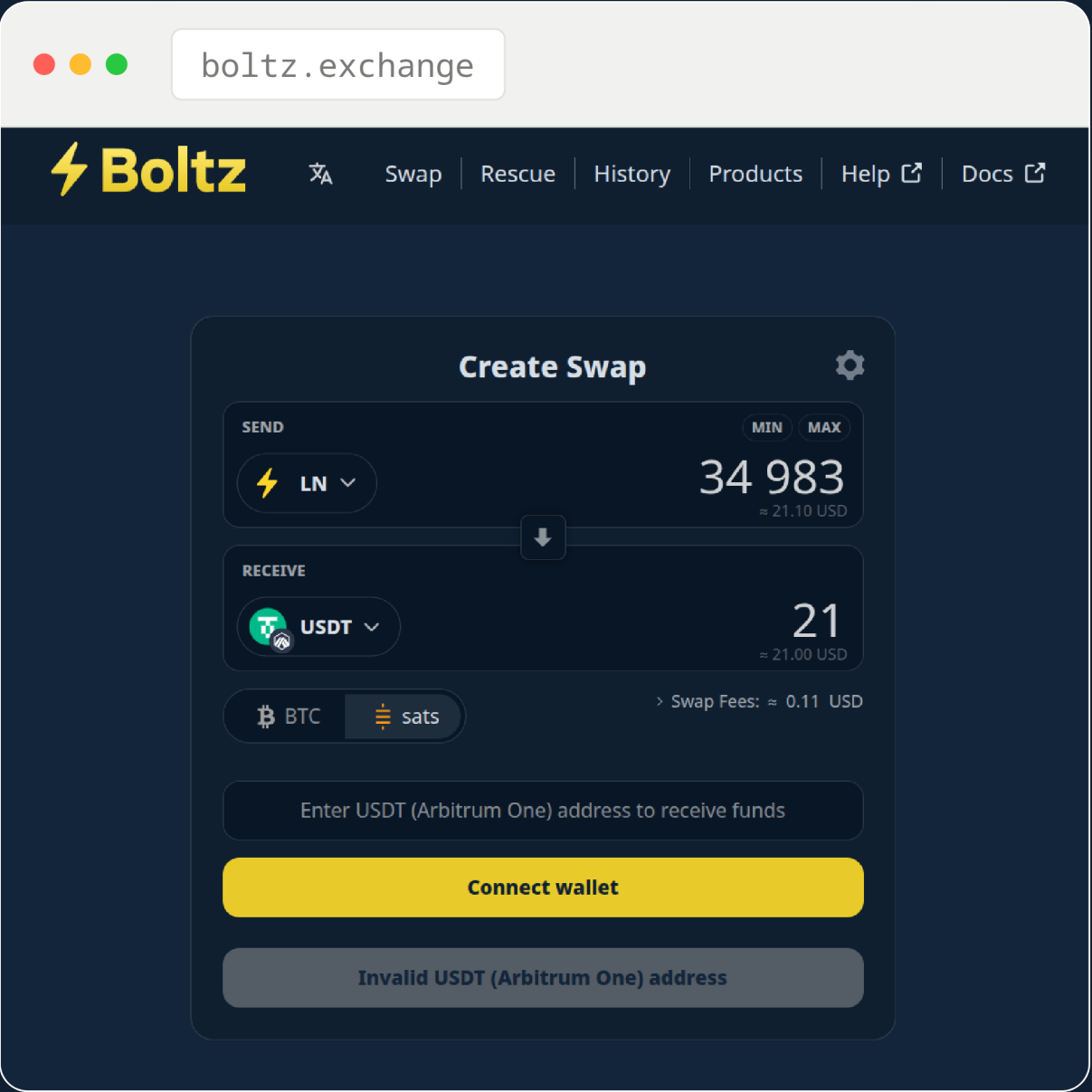
**Build Things
People Use.**

Then bring them home to Bitcoin.

TRY IT · BREAK IT · BUILD ON IT
PLAN ₿ SUMMER SCHOOL · LUGANO

LIVE IN PRODUCTION · OPEN SOURCE

boltz.exchange



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